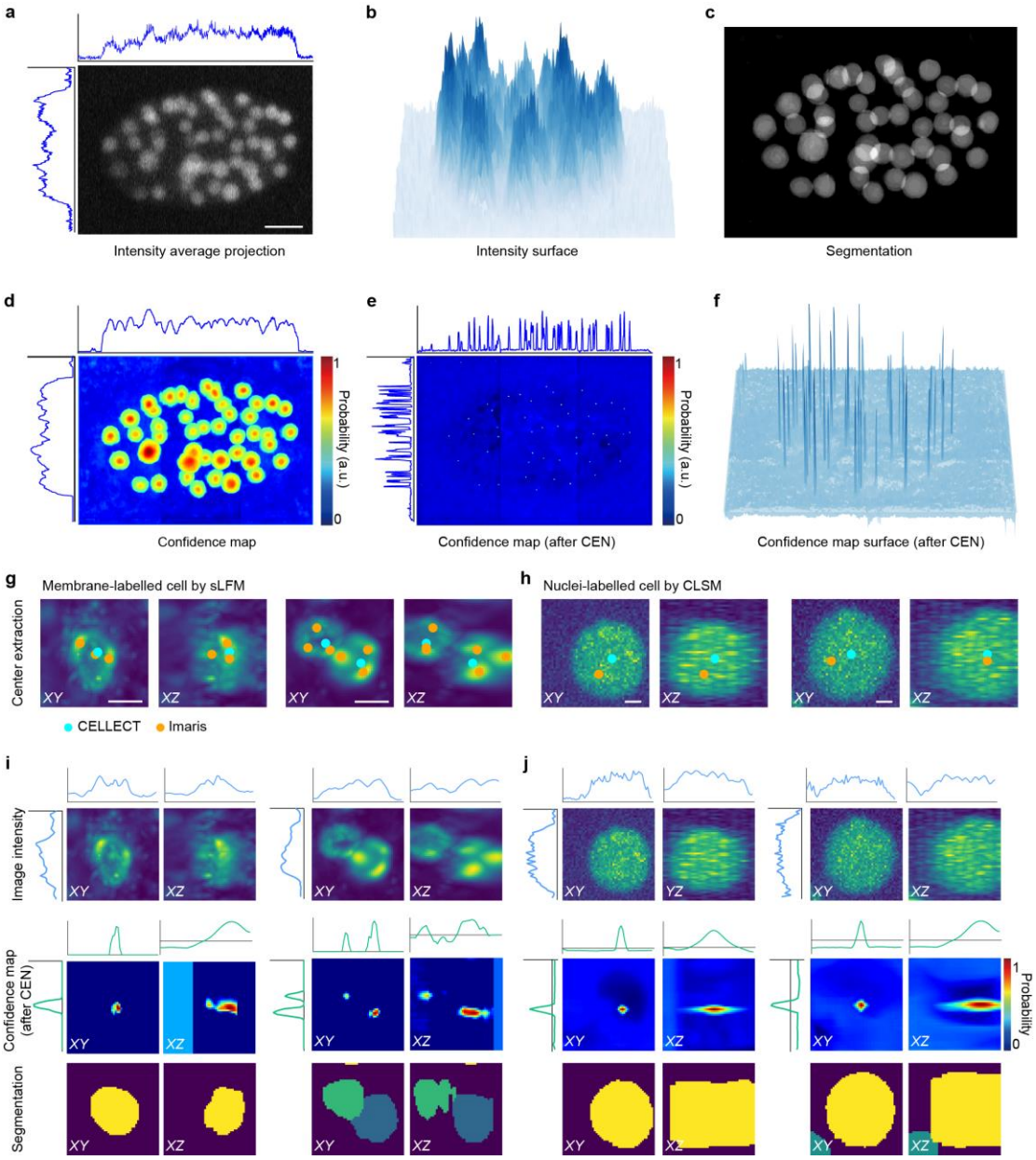


Supplementary Figure 2 | Examples of the output confidence map of cell centers obtained by CELLECT. a-b, Visualization of the intensity distribution of the example data. (**a**) 2D average projected image of the intensity distribution of input image. (**b**) 3D visualization of the intensity surface. **c**, Cell segmentation regions generated by the CELLECT model. **d-e**, 2D maximum projected image of the confidence map (**d**) before and (**e**) after CEN. **f**, 3D visualization of the confidence map surface after CEN. **g-h**, Comparison of center extraction performance between CELLECT and Imaris for (**g**) membrane-labeled cells image acquired by using Scanning Light-Field Microscopy (sLFM) and (**h**) nuclei-labeled cells imaged using Confocal Laser Scanning Microscopy (CLSM). **i-j**, Input image intensity, confidence maps (after CEN), and segmentation results produced by the CELLECT model. Panels (**a–f**) use data from the Cell Tracking Challenge (CTC) Fluo-N3DH-CE dataset. Panels (**g–j**) use nuclei-labeled data from the mskcc-confocal dataset and membrane-labeled neutrophil data from mouse imaging, as also shown in **Fig. 4**. All